Chem 150 **Term Test 2A** **28 March 2024** **Name of student** _____

1.(**2 marks**) Write a dissociation equation for K_2SO_4 in water.

$K_2SO_4 \leftrightarrow 2K^+ + SO_4^{2-}$

2.(**2 marks**) Write an ionization equation for HNO_3 in water.

$HNO_3 + H_2O \rightarrow NO_3^- + H_3O^+$

3. (**2 marks**) Sodium chloride dissolves spontaneously in water at room temperature. The standard enthalpy for this process NaCl(s) →NaCl(aq) is + 5 kJ/mol. What can you confidently say about the standard entropy for this process?

@ 25 °C :

$\Delta G = \Delta H - T\Delta S$

$- \quad + \quad +$

$(298\ K)(\Delta S) > 5\ kJ/mol$

$\Delta S > \frac{5\ kJ/mol}{298\ K} > 16.78\ J/mol \cdot K$

4. (**3 marks**) Calculate the pH of 0.00112 M $Ca(OH)_2$(aq)

$Ca(OH)_2 + H_2O \rightleftharpoons Ca^{2+} + 2OH^-$

$pOH = -\log(2 \cdot 0.00112) = 2.6498$

$pH = 14 - 2.6498 = \boxed{11.4}$

5. (**3 marks**) As far as thermodynamics is concerned define a closed system. Write a sentence! Next give an example of a closed system.

A closed system allows for the exchange of energy but not matter between the system and the surroundings,

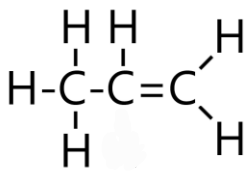
eg. a water bottel

6.(**6 marks**) Calculate the enthalpy of combustion of propene compound A and cyclopropane compound B using the thermodynamic data given in the formulas and constants section. The final result should have the unit kJ/mol and should be based on the heat produced per mole of compound combusted. The two enthalpies for the isomers are different. Which compound is more stable and how can you tell?

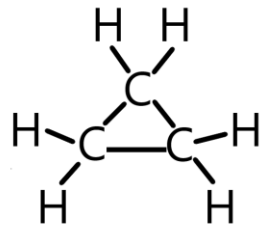
$CO_2 = -393.5$

$H_2O = -285.8$

C_3H_7



Compound A



Compound B

C_3H_6

20.42 kJ/mol

53.3 kJ/mol

Compound A :

$$4C_3H_7 + 19O_2 \rightarrow 12CO_2 + 14H_2O$$

$$12(-393.5\,kJ/mol) + 14(-285.8\,kJ/mol) - [4(20.42\,kJ/mol)]$$

$$= -8804.9\,kJ/4\,mol = -2201.2\,kJ/mol$$

Compound A

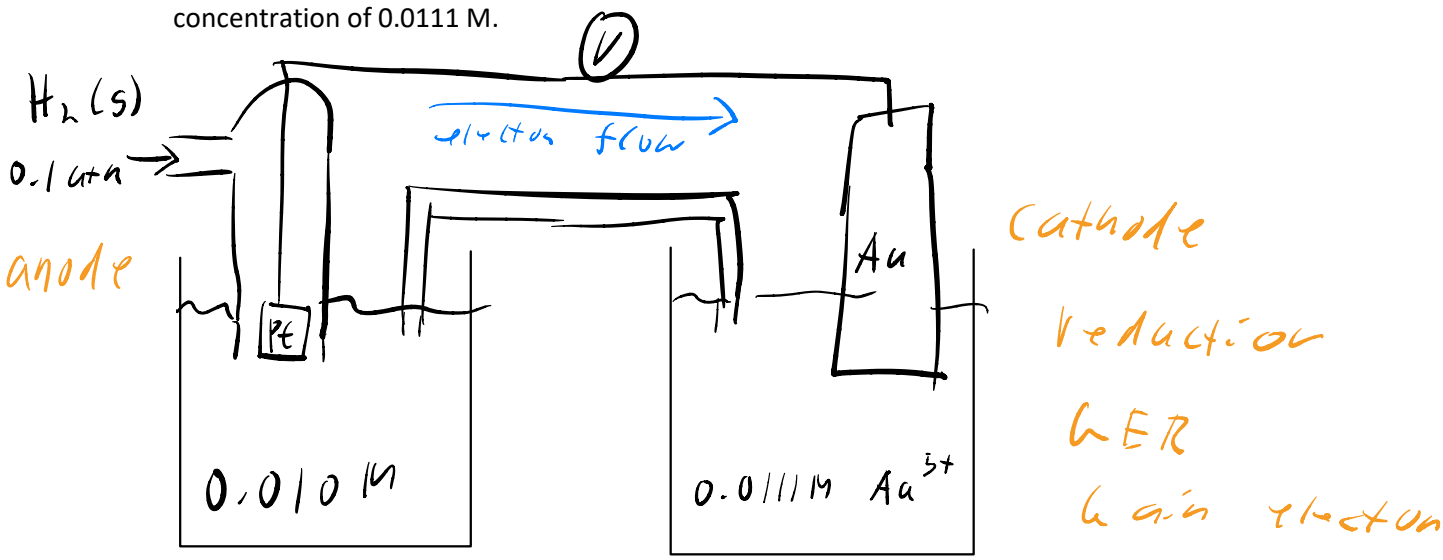
$$12(-393.5\,kJ/mol) + 14(-285.8\,kJ/mol) - [4(53.3\,kJ/mol)]$$

$$= -9018.1\,kJ/4\,mol = -2254.5\,kJ/mol$$

Compound B

Compound A is more stable since its enthalpy of combustion is smaller (release less energy during combustion)

7. (**10 marks**) Draw a hydrogen electrode properly connected to a gold half cell. Name all vital parts of this Voltaic cell in your drawing. The hydrogen ion concentration in the hydrogen electrode is 0.010 M and the $H_2(g)$ pressure is 0.1 atm. The gold electrode is dipping into a solution with a gold(III) ion concentration of 0.0111 M.



Which electrode is the Anode and which electrode is the Cathode in your drawing?

Give the half reactions that occur on both electrodes and give the overall cell reaction.

Cathode half reaction $2\,[Au^{3+}(aq) + 3e^- \rightarrow Au(s)]$ $E_{cell} = 1.52$

Anode half reaction $3\,[H_2(s) \rightarrow 2H^+ + 2e^-]$

Overall cell reaction

$$2Au^{3+}(aq) + 3H_2(s) \rightarrow 2Au(s) + 6H^+$$

Calculate the cell potential E_{cell} $E_{cell}^{\circ} = 1.52\ V$

$$2Au^{3+}(0.0111\ M) + 3H_2(0.1) \rightarrow 2Au(1) + 6H^+(0.01M)$$

$$Q = \frac{[1]^2[0.01]^6}{[0.0111]^2[0.1]^3} = 8.1162 \times 10^{-6}$$

$$E_{cell} = 1.52 - \frac{0.0592\ V}{6} \log(8.1162 \times 10^{-6}) = \boxed{1.57\ V}$$

Draw an arrow in your cell diagram indicating the direction flow of electrons

8. (**8 marks**) Calculate the pH of 0.111 M aqueous sodium nitrite: $NaNO_2(aq)$. The acid ionization constant for nitrous acid (HNO_2) is $\mathbf{K_a = 7.2 \times 10^{-4}}$

$$NaNO_2(aq) \rightarrow \underbrace{Na^+}_{\text{strong base ion}} + \underbrace{NO_2^-}_{\text{weak acid ion}}$$

R:

$$NO_2^- + H_2O \leftrightarrow HNO_2 + OH^-$$

I:	0.111 M	0 M	0 M
C	$-x$ M	$+x$ M	$+x$ M
E	$0.111 - x$ M	x M	x M

$$K_b = \frac{K_w}{K_a} = \frac{1 \times 10^{-14}}{7.2 \times 10^{-4}} = 13.89 \times 10^{-12}$$

assuming $x << 0.111$

$$13.89 \times 10^{-12} = \frac{x^2}{0.111 - x}$$

$$13.89 \times 10^{-12} = \frac{x^2}{0.111} \qquad x = \sqrt{(13.89 \times 10^{-12})(0.111)}$$

$$= 1.2416 \times 10^{-6}$$

(Yes much smaller than 0.111)

$$pOH = -\log(1.2416 \times 10^{-6})$$

$$= 5.906$$

$$pH = 14 - 5.906 = 8.094$$

Formulas and constants:

Selected standard electrode half potentials:

$Au^{3+}(aq) + 3e^- \rightarrow Au(s)$	**+1.52 V**
$O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$	**1.229 V**
$Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$	**+0.771 V**
$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$	**+ 0.340 V**
$2H^+(aq) + 2e^- \rightarrow H_2(g)$	**0 V (as defined)**
$Mo^{3+}(aq) + 3e^- \rightarrow Mo(s)$	**- 0.20 V**
$Cr^{3+}(aq) + e^- \rightarrow Cr^{2+}(aq)$	**- 0.424 V**
$Fe^{2+}(aq) + 2e^- \rightarrow Fe(s)$	**- 0.440 V**
$Mn^{2+}(aq) + 2e^- \rightarrow Mn(s)$	**- 1.18 V**
$Mg^{2+}(aq) + 2e^- \rightarrow Mg(s)$	**- 2.356 V**

Some useful equations: **$S = k\ln W$, $k = 1.38064852 \times 10^{-23}\ JK^{-1}$**

$\Delta H° = \Sigma v_p \Delta H_f°(products) - \Sigma v_r \Delta H_f°(reactants)$ **$\Delta G° = \Delta H° - T\Delta S°$**

Selected standard enthalpy of formations:

CH_3NO_2 (l)	**– 112.6 kJ/mol**
Cyclopropane C_3H_6 (g)	**+ 53.3 kJ/mol**
propene C_3H_6 (g)	**+ 20.42 kJ/mol**
CO_2(g)	**- 393.5 kJ/mol**
H_2O(l)	**-285.8 kJ/mol**

$E°_{cell} = E°_{reduction\ half\ cell} - E°_{oxidation\ half\ cell}$ (where E° refers to the reduction half potentials)

$E_{cell} = E_{cell}° - \frac{0.0592\ V}{n} \log Q$

$pH = -\log[H_3O^+] = -\log[H^+]$ **$pOH = -\log[OH^-]$** **$pH + pOH = 14 = pK_w$**

For an acid HA that ionizes in water to give H_3O^+ and A^- the acid constant K_a is defined as:

$K_a = \frac{[H_3O^+][A^-]}{[HA]}$ For an acid and its conjugated base (or a base and its conjugated acid) in water: **$K_a \times K_b = K_w = 10^{-14}$**